CBAM AND EXPORT COMPETITIVENESS: MARKET SHARE ELASTICITY ESTIMATES AND SIMULATIONS FOR BRAZIL AND ARGENTINA[1]*

CBAM E COMPETITIVIDADE DE EXPORTAÇÕES: ESTIMAÇÃO DE ELASTICIDADES DE PARTICIPAÇÃO DE MERCADO E SIMULAÇÕES PARA BRASIL E ARGENTINA

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ABSTRACT

This study analyzes the potential effects of the European Union (EU) Carbon Border Adjustment Mechanism (CBAM) on the competitiveness of Brazilian and Argentine exports in the European market. Using a structural gravity model estimated via PPML with fixed effects, market share price elasticities are estimated at the individual product level for aluminum, iron and steel, and cement, covering trade flows from over 150 countries to the EU between 2010 and 2025, combined with CBAM cost simulations under three carbon price scenarios and two tariff regimes. Results indicate that impacts are negative but moderate and highly heterogeneous across products: in iron and steel, Brazil's exposure concentrates in higher value-added products, and for Argentina, lower substitution elasticities mitigate the competitive impact. In aluminum, Mercosur tariff elimination offsets the CBAM burden under lower carbon price scenarios. The central contribution lies in demonstrating that product-level disaggregation is decisive for assessing CBAM exposure, as sector-level analyses systematically obscure the heterogeneity that determines the actual competitive impact on each country's export portfolio.

Keywords: Carbon Border Adjustment Mechanism. Export competitiveness. Gravity model. Market share elasticity. Mercosur and EU free trade agreement.

RESUMO

Este estudo analisa os potenciais efeitos do Mecanismo de Ajuste de Carbono na Fronteira (CBAM) da União Europeia sobre a competitividade das exportações brasileiras e argentinas no mercado europeu. Por meio de um modelo gravitacional estrutural estimado via PPML com efeitos fixos, são estimadas elasticidades-preço de participação de mercado ao nível de produto individual nos setores de alumínio, ferro e aço, e cimento, cobrindo fluxos comerciais de mais de 150 países para a UE entre 2010 e 2025, combinados com simulações de custo do CBAM em três cenários de preço de carbono e dois regimes tarifários. Os resultados indicam impactos negativos, porém moderados e heterogêneos entre produtos: no ferro e aço, a maior exposição do Brasil concentra-se nos produtos de maior valor agregado, já para a Argentina, menores elasticidades de substituição atenuam o impacto. Para o alumínio, a eliminação tarifária do acordo Mercosul-UE é suficiente para compensar o custo do CBAM nos cenários de menor preço de carbono. A contribuição central está em demonstrar que a desagregação ao nível de produto é decisiva para avaliar a exposição ao CBAM, uma vez que análises setoriais agregadas obscurecem a heterogeneidade que determina o impacto competitivo efetivo sobre o portfólio exportador de cada país.

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Palavras-chave: Mecanismo de Ajuste de Carbono na Fronteira. Competitividade de exportação. Modelo gravitacional. Elasticidade de participação de mercado. Acordo Mercosul e União Europeia

1 INTRODUCTION

The EU's growing climate ambitions have brought about profound changes in the regulatory environment for international trade, with direct impacts on the competitiveness of exporters from developing countries. At the heart of these changes lies the Carbon Border Adjustment Mechanism (CBAM), established by Regulation (EU) 2023/956 as part of the Fit for 55 package and the 2019 European Green Deal. The mechanism operates by imposing additional costs on imports of carbon-intensive goods based on their embedded emissions content, thereby equalizing the carbon price between European producers subject to the EU Emissions Trading System and foreign exporters. By introducing production carbon intensity as a determining variable of relative price in the European market, the CBAM reshapes the conditions of international competition in the covered sectors, making exporters' market share sensitive not only to traditional production efficiency but also to the emissions profile of each origin (Fontagné; Schubert, 2023; European Union, 2021).

In this context, Brazil and Argentina hold a position of particular strategic importance. As the two largest EU trading partners of the Mercosur, with trade volumes of €87.1 billion and €17.4 billion, respectively, in 2025 (Eurostat, 2026a). Both countries have significant exports in the sectors covered by the CBAM and distinct production and emissions profiles, which implies differing exposures to the mechanism. Additionally, the ongoing negotiation of the Mercosur-EU free trade agreement adds a layer of complexity to the analysis, as potential tariff reductions could act as a partial counterweight to the costs imposed by the CBAM, altering the competitive balance between the bloc's exporters and their global competitors. Understanding the magnitude of these effects on Brazilian and Argentine exports thus becomes a matter of both academic and trade policy relevance (Grand; Schulz-Antipa; Rozenberg, 2023; Campos *et al.*, 2026).

The existing literature has made significant progress in analyzing CBAM effects on exports, GDP, and welfare using computable general equilibrium and structural gravity models. However, most analyses capture aggregate macroeconomic effects without identifying market share redistribution among competing exporters, precisely the channel through which the CBAM operates, making market share price elasticities the central parameter for assessing its competitive effects (Dechezleprêtre *et al.*, 2025). This gap remains largely unexplored for Brazil and Argentina.

Thus, the research question of this study is: **What are the potential effects of the EU's Carbon Border Adjustment Mechanism (CBAM) on the relative competitiveness of Brazilian and Argentine exports in the European market, given the changes in relative prices and market shares resulting from carbon pricing?** The overall objective is to analyze the substitution effects induced by the CBAM on Brazilian and Argentine exports in the European market by estimating product-level market-share price elasticities and simulating relative price changes under alternative carbon-price and tariff scenarios. The central contribution of this work lies in the econometric estimation of specific market share elasticities for Brazilian and Argentine products in the sectors covered by the CBAM, an approach that complements the existing literature by offering a perspective focused on redistributive effects among competing exporters, providing insights directly applicable to the formulation of trade policy strategies for both countries in the context of the global climate transition.

Finally, the article is structured as follows: Section 2 develops the theoretical framework, covering the fundamentals of CBAM, its economic logic, and the existing empirical evidence for Brazil and Argentina. Section 3 presents the methodology, describing the data used, the specification of the gravity model estimated via PPML, and the simulation framework. Section 4 reports the results, beginning with the analysis of the estimated elasticities and the explanatory power of the model, followed by simulations of the impact of CBAM on the iron and steel and aluminum sectors for Brazil and Argentina. Section 5 presents the final considerations. Claude AI was used for writing revision (Annex B).

2 THEORETICAL FRAMEWORK

This section aims to develop the conceptual framework underlying the empirical analysis proposed in this study. It begins by presenting the Carbon Border Adjustment Mechanism, passing by its origins within the European climate policy agenda, the objectives the EU seeks to achieve through its implementation, and the two impact channels identified in the literature. It then examines the mechanism from an economic perspective, connecting it to the theory of externalities, international trade economics, and the microeconomic toolkit for price and demand analysis, with particular emphasis on the role of elasticities as the central parameter for measuring its competitive effects. Finally, it reviews the existing empirical evidence on the impacts of the CBAM on developing economies, with a focus on the Brazilian and Argentine cases. Consensus AI was used for references gathering as presented in Annex B.

2.1 THE CBAM

The Carbon Border Adjustment Mechanism (CBAM) was initially proposed in 2021 as part of the Fit for 55 package that established climate-related laws regarding lowering greenhouse gas emissions to tackle the objectives of the European Green Deal Act of 2019. The Green Deal, launched by the European Commission in December 2019, established a roadmap for the EU to become carbon neutral by 2050, setting an interim target of reducing emissions by at least 55% compared to 1990 levels by 2030 (European Commission, 2019). In this context, the Commission identified carbon leakage as an expressive threat to the effectiveness of its internal climate policy, prompting the development of a border adjustment mechanism (Simões, 2023). Following a formal legislative proposal in July 2021, the mechanism was approved in April 2023 and entered into force through Regulation (EU) 2023/956. A transition phase took place between October 2023 and December 2025, during which importers reported embedded emissions without financial obligations. The definitive regime, with full requirements for the purchase of certificates, began on January 1, 2026 (European Union, 2023b).

The EU formally justifies the CBAM on the basis of two interrelated grounds. The first is the prevention of carbon leakage, the risk that stricter domestic carbon pricing under the EU ETS would encourage the relocation of carbon-intensive production to countries with less stringent climate regulations, undermining European decarbonization efforts without reducing global emissions (European Union, 2023b). The second is the restoration of a level playing field between European producers subject to the ETS and importers from countries where no equivalent carbon cost is applied, ensuring that the competitiveness of European industry is not eroded by regulatory asymmetries (Dechezleprêtre *et al.*, 2025). The mechanism operates by requiring European importers of covered goods to purchase CBAM certificates corresponding to the carbon price that would have been paid had those goods been produced under the EU ETS. Importers may deduct any carbon price already paid in the

country of origin, thereby avoiding double taxation on emissions already priced domestically. The covered sectors are iron and steel, aluminium, cement, fertilisers, hydrogen, and electricity, and were selected based on their carbon intensity and exposure to international competition, as defined in Annex I of the Regulation (European Union, 2023b).

Literature identifies two main expected outcomes of the CBAM. The first is a shift in the import portfolio: by raising the effective price of goods from carbon-intensive exporters relative to cleaner alternatives, the mechanism is expected to redirect European demand for imports toward countries with lower carbon intensity in production (Larch; Wanner, 2017). The second is the incentive for technological change in exporting countries: by internalizing the cost of carbon into export prices, the CBAM creates economic pressure for exporters to invest in greener production processes, effectively extending the reach of carbon pricing beyond European borders (Clausing *et al.*, 2025). These two channels are conceptually distinct and operate over different time horizons. The trade reallocation effect is expected to materialize relatively quickly, as importers respond to changes in relative prices. The technological change effect, in turn, operates over a longer horizon, as it requires exporters to make structural investments in production processes. This study focuses on the first channel, estimating the extent to which price variations induced by the CBAM alter the competitive positioning of Brazilian and Argentine exporters in the European market for aluminum, iron and steel, and cement.

Lastly, distributional impacts of the CBAM are likely to vary among exporting countries. Economies with more carbon-intensive production processes and greater dependence on the European market face greater exposure to the cost of the mechanism (UNCTAD, 2021). Developing countries are particularly vulnerable in two aspects: their industrial sectors tend to rely on older, less efficient technologies, resulting in higher embedded emissions intensities, and they often lack the carbon accounting infrastructure and domestic pricing systems necessary to claim deductions under Article 9 of the Regulation (Grand; Schulz-Antipa; Rozenberg, 2023; Morone; Alfino, 2025). This combination of higher costs and limited offsetting capacity places exporters from developing countries at a structural disadvantage relative to competitors from countries with established carbon markets.

2.2 THE ECONOMICS OF CBAM

The CBAM can essentially be understood as a corrective tax on a negative externality. In the tradition of Pigou (1932), the optimal response to an externality is the imposition of a tax equal to the marginal social cost of the activity that generates it, in this case, the carbon emissions embedded in the production of traded goods. The case for such a tax was reinforced by Stern (2007), who characterized climate change as the greatest market failure in history, resulting from the inability of markets to price the social cost of greenhouse gas emissions. Without intervention, producers in countries without carbon pricing face lower costs than their counterparts in regulated economies, not because they are more efficient, but because they externalize environmental costs that are borne globally. However, the CBAM is not a classic Pigouvian tax in its purest form. A standard Pigouvian instrument would tax emitting activity directly at the source. The CBAM, by contrast, operates at the border, taxing the importer based on the carbon content of the good, rather than the producer for the act of emitting (European Union, 2023b). This design is motivated by the problem of carbon leakage: in an open economy, a domestic carbon tax without border adjustment creates an incentive for production to relocate to unregulated jurisdictions, reducing domestic emissions while increasing global emissions, a phenomenon widely documented in the literature (Cameron; Baudry, 2023; Grubb *et al.*, 2022). The CBAM aims to extend the logic of

Pigouvian correction to the margin of trade, ensuring that the cost of carbon is applied regardless of the location of production.

Within the framework of international trade theory, a nation's competitive advantage is determined by its relative production costs, a principle formalized by Ricardo (1821) in his theory of comparative advantage. When environmental regulations are absent or weak, carbon-intensive producers in unregulated economies face systematically lower production costs, not as a result of genuine productive efficiency, but because of the implicit subsidy created by the absence of carbon pricing. This dynamic is captured by the pollution haven hypothesis, which argues that environmental regulatory asymmetry generates artificial comparative advantages, encouraging the concentration of carbon-intensive industries in less-regulated jurisdictions (Copeland; Taylor, 2003). In this sense, the CBAM can be interpreted as an attempt to correct a distortion in the international pricing system, ensuring that the cost of carbon is reflected in traded goods regardless of the place of production. Countries with lower emission intensities, whether due to cleaner technologies, less carbon-intensive energy mixes, or domestic carbon pricing, face a lower cost burden and are therefore relatively advantaged in accessing the European market (UNCTAD, 2021). This introduces an environmental dimension into the classical Ricardian and neoclassical Heckcher-Ohlin framework: comparative advantage is no longer determined exclusively by factor endowments and productivity, but also incorporates the carbon intensity of production. There is, however, an inherent tension in this design: the line between a legitimate environmental correction and a protectionist measure is legally and economically contested, raising questions about its compatibility with WTO rules (Pauwelyn, 2013).

At the microeconomic level, the CBAM functions as a differentiated tax on imported goods, varying according to the embedded carbon content of each product and its country of origin. This differentiation introduces asymmetric variations in the relative prices faced by European importers among competing suppliers, a mechanism whose effects can be analyzed using the standard framework of consumer theory. When the price of a good from a given origin increases relative to substitutes, two effects emerge: an income effect, reflecting the reduction in importers' real purchasing power, and a substitution effect, reflecting the reallocation of demand toward relatively cheaper alternatives (Varian, 2014). In the context of the CBAM, it is the substitution effect that takes center stage analytically, as importers face higher costs for goods from high-emission, demand is expected to shift toward lower-emission sources, altering the competitive positioning of exporters in the European market. The magnitude of the substitution effect is not uniform, it depends fundamentally on the price elasticity of demand by origin. The greater the sensitivity of European importers to relative price changes among suppliers, the greater the redistribution of market share induced by the CBAM. This relationship makes the price elasticity of demand for imports the central parameter for quantifying the competitive impact of the mechanism (Dechezleprêtre *et al.*, 2025). In practical terms, a carbon-intensive exporter will face a relative price increase, but the extent to which this translates into a loss of market share depends entirely on how elastic European demand for its products is relative to available alternatives. The empirical literature on international trade documents wide heterogeneity in the estimated elasticities across sectors and countries of origin (Yotov *et al.*, 2016). This heterogeneity has direct implications for the CBAM analysis: exporters within the same sector may face very different competitive impacts depending on their specific elasticities, which makes the use of aggregate values from the literature inappropriate for evaluating particular cases. In the context of this study, Brazil and Argentina operate in sectors with their own competitive dynamics and distinct positions in the European market, which justifies the estimation of country and product-specific elasticities as a necessary step for simulating the impacts of the CBAM on their competitiveness.

2.3 RELATED STUDIES

The study by Feijó and Kuik (2016) sought to explain how Brazil's environmental vulnerability would affect the competitiveness of its industry if the EU were to implement Border Carbon Adjustments (BCAs). To this end, the authors used the GTAP-EP computable general equilibrium model, a modified version of GTAP-E that innovates by integrating emissions generated by industrial processes and the indirect costs resulting from increased electricity and gas prices under carbon taxation. The results indicated that BCAs would negatively affect Brazilian exports from sectors such as non-metallic minerals and iron and steel. Specifically, the simulation indicated that the *ad valorem* tariffs applied to Brazil could reach 6.45% for iron and steel and 5.97% for non-metallic minerals in the most stringent scenario, resulting in a loss of competitiveness relative to China and India in the latter sector.

Korpar, Larch and Stöllinger (2022) aimed to estimate the CBAM effects on exports, real GDP, and overall well-being. The methodology employed was a multisectoral and multiregional structural gravity model, which allowed for the calculation of equivalent carbon tariffs based on the emissions intensity of 128 regions. When analyzing the impact on South America, the study concluded that the macroeconomic effects are modest but negative for the region's exporters: Brazil's total exports would decline by 0.21%, while Argentina's would fall by 0.05%.

Focusing on the Asia-Pacific region, the Mortha, Arimura, Takeda and Chesnokova (2023) study simulated the impact of the CBAM on trade and production, providing global data for comparison. The study used a structural gravity model based on the Anderson-Armington CES[3] production function, innovating by including in the calculation the quantification of emissions from international transport and maritime freight activities. For South America (SAM), the results indicated a significant reduction in exports from intensive sectors: exports of iron and steel would fall by 2.24%, and those of non-ferrous metals would decrease by 0.57%. Despite this, the study demonstrated that the total reduction in emissions in the region would amount to 3.72 million tons of CO2, driven mainly by the decrease in long-distance transport.

Finally, the article by Perdana, Vielle and Oliveira (2023) assessed the implications of the CBAM specifically for energy-intensive Brazilian industries, analyzing scenarios for Brazil's integration into "climate clubs." The approach used the GEMINI-E3 dynamic general equilibrium model, simulating impacts based on different scopes of carbon content (direct and indirect from electricity). Contrary to initial concerns, the study concluded that the CBAM could improve the trade balance of Brazil's energy-intensive sectors by $10.7 billion by 2040 under Scope 2[4], as the country's clean electricity mix gives its products a lower CO2 content compared to global competitors. Production in these sectors in Brazil would increase by 1.8%, creating approximately 74,000 new jobs.

3 METHODOLOGY

In order to assess the CBAM's possible impacts on Brazilian and Argentinian competitiveness of iron and steel, aluminium and cement products regarding the EU market, this research follows a two-step methodology. At first, a modelling step focused on elasticity

[3] The Gravity Model based on the Anderson–Armington CES is an international trade framework derived from CES (Constant Elasticity of Substitution) preferences, in which consumers substitute goods from different origins at a constant rate, and from the Armington assumption that products are differentiated by country of origin, allowing bilateral trade flows to be explained by economic size, trade costs, and relative competitiveness.
[4] Considering both direct emission from the production process (scope 1) and emission from the energy used for the production process (scope 2).

estimation, followed by a CBAM simulation step. In the first step, we use historical trade data to estimate the EU price elasticity of import demand. This step aims to capture how variations in relative prices affect the allocation of market shares among exporting countries. In the simulation step, we use the emission factors to estimate the import price variation by simulating the additional cost associated with embedded carbon emissions. This step allows for the assessment of how the mechanism alters relative prices across exporting countries and, consequently, their competitive positioning in the European market. Finally, the two components are combined by applying the simulated price changes to the estimated elasticities, allowing for the projection of the potential impact of the CBAM on demand and competitiveness.

In terms of methodological classification, this research is characterized, regarding the approach to the problem, as quantitative, since it draws on structured numerical data and econometric techniques to measure relationships between variables and produce inferences about the CBAM's impacts on export competitiveness. Regarding its objectives, it is classified as descriptive and explanatory, as it simultaneously characterizes the competitive positioning of Brazil and Argentina in the European market and seeks to identify the causal relationships between relative price variations, induced by the carbon mechanism, and changes in market shares. Regarding technical procedures, it is configured as documentary, given that it is grounded in secondary data extracted from official databases that will be further exposed, upon which a descriptive and econometric analysis is applied, combining elasticity estimation via PPML with fixed effects and simulation of climate and trade policy scenarios.

3.1 THE DATA

The data used in this analysis were divided into three categories that will be explained separately. The first category is trade data, which is the base for the modelling and estimation as it was used both to estimate the import prices and the shares. The second is import tariff data, which was used both to approximate the import prices to the effective paid price for EU importers and to build the Mercosur-EU agreement scenario. And the third is the emission factor, which was used to estimate the CBAM impact on price.

3.1.1 Trade data

All trade data were extracted from the EU official database, EUROSTAT, in April 2026. In this case, the database “International trade of EU and non-EU countries since 2002 by HS2-4-6” was used as the data source. Even though there is data availability from 2002, the analysis covers yearly data from 2010 to 2025. This reduced time scope is due to the conjunctural objectives of the analysis, as well as the intention of ensuring that punctually relevant events like the 2008 crisis would not have an impact on the model.

Within the scope of the study, the trade data were structured as combinations of exporter country, product and year. In total, 62 main products were analysed across the three sectors, following Annex I of the Regulation (EU) 2023/956. Products were identified by their Harmonised System (HS) codes, aggregated at the HS4 level for most categories, with specific products referenced in the Regulation disaggregated to the HS6 level. At an exporter level, data from more than 200 countries and territories were available. Intra-EU trade flows were also retained in the dataset, as shifts in internal bloc commerce represent a potential outcome of CBAM-induced trade reallocation.

The data extracted from this database were the volume of trade, measured in net kilograms of imports, and value, measured in euros imported. This data was used to build the price measured in euros per ton imported and the shares, based on the volume imported. The

prices consider Cost, Insurance and Freight (CIF) Incoterm, which endogenises factors like geographic distance, considered in the freight component of the price, as well as some institutional risk considered in the insurance component. Those would count as the gravitational model "frictions" other than the tariffs (Yotov *et al.*, 2016).

The market share is the dependent variable in this study and was constructed by dividing the imported volume of a specific product from a specific exporter by the total EU imported volume of that product in a given year. According to Yotov *et al.* (2016), the use of market share is appropriate because it normalizes trade flows by the importer's total expenditure and allows us to isolate the actual response in volume to direct price-displacers, such as tariffs and taxes. In that case, our base model aligns with the theoretical framework of structural gravity (Armington-CES), where an exporter's market share depends on its price relative to the average ease of access of all competitors to the global market. These trade data constitute the foundation of both the econometric estimation and the CBAM simulation, with complementary data on import tariffs and emission factors described in the following subsections.

3.1.2 Tariff data

EU import tariff data were extracted from the Market Access Map (MAcMap) tool, developed by the International Trade Centre, in May 2026. In this case, the data were also structured as a combination of exporter country, product, and year. The data followed the effective *ad valorem* tariff applied by the country, which was considered the lowest tariff rate available for the importer, whether from specific trade agreements, preference systems, or the base tariff known as the Most-Favoured-Nation (MFN) tariff. For products defined at the HS4 level, the tariff applied corresponds to the simple average of all constituent HS6 codes.

This data was used to approximate the price calculated from the trade data to the actual effective price paid by EU importers, as the CIF price does not cover the import taxation cost. It was also used for the construction of the Mercosur-EU free trade agreement scenario that will be presented in subsection 3.3. It is worth noting that tariff data availability varies across exporters and years, a limitation that is addressed in the sample construction described in subsection 3.1.4.

3.1.3 Emission factors

Finally, the emission factors were the impact variable, used to estimate the CBAM cost impact, representing the shock applied to the model developed with the trade and tariff data. An emission factor is basically an estimation of how much greenhouse gas is emitted with the production of a certain good. In this case, we are following the EU data format, which is tons of carbon dioxide equivalent (CO_2e) emissions per tonne of goods.

The emission factor data used in the analysis were collected from Annex I of the Regulation (EU) 2025/2620. These default values were determined on the basis of data collected during the CBAM transitional period (2023–2025), and are specific to each product and country of origin for the main exporting countries. For exporters not covered by country-specific estimates, the regulation provides a fallback default value, ensuring that default values are not underestimated. In practice, the default value applies when the importer is unable or opts not to submit verified installation-level emission reports, but is also an approximation of real country-specific emissions.

An important feature of these default values is the inclusion of a mark-up, applied on top of the 2025 baseline emission values, that, according to the EU, is to ensure that emissions will not be underestimated. This mark-up is structured as a gradual annual increment of 10

percentage points: 10% in 2026 to 30% in 2028. A comprehensive review of the entire system is foreseen for 2027, with adjustments to be made based on experience accumulated during the transitional period.

The default values are presented at the Combined Nomenclature 8-digit level (CN8), which corresponds to the HS6 classification with two additional digits of national specificity. Where the level of disaggregation in the regulation was finer than the one adopted in this analysis, a simple average of all constituent components was applied: for products analyzed at the HS6 level, the emission factor corresponds to the simple average of all CN8 codes within that HS6; for products analyzed at the HS4 level, the emission factor corresponds to the simple average of all constituent HS6 codes within that HS4. These emission factors, combined with the carbon price scenarios, constitute the basis of the CBAM cost simulation described in subsection 3.3.

3.1.4 Sample construction

The sample passed through 4 significant filters before being used both in the modelling and simulation phases. The first filter was directly in the download of the trade data, as the observations with no data were excluded from the sample by the EUROSTAT download tool. Already within the analysis tool (RStudio), countries with no tariff data were excluded, as the absence of tariff data prevents the construction of the effective price variable, which is central to the estimation. Additionally, there was a filter for prices equal to or lower than zero, as this would count for statistical mistakes in the dataset.

The first methodological filter was regarding a minimum amount exported. In this case, observations with less than 1 ton exported were excluded from the analysis, as this would generate exaggerated prices that could have an artificial impact in the modelling with no real volume significance. The second filter was related to ensuring a minimum import consistency, considering only countries that exported at least 1 ton of the analysed product for at least 5 years to the EU. This filter aims to both exclude extreme punctual exports and to ensure that there will not be any singleton[5] in the sample. The result of these filters can be seen in Table 1 below.

Table 1 – Data analysed

Sector	CN Codes	Observations	Exporters	% of total volume imported
Aluminium	14	13,113	114	99.6%
Cement	6	4,104	81	99.8%
Iron and Steel	42	38,726	159	99.5%
Total	62	55,943	-	-

Source: Compiled from data obtained in the research.

Despite the successive filters applied, the final sample retains over 99% of the total volume imported by the EU in each sector, confirming the representativeness of the estimation sample in relation to the universe of trade flows.

3.2 MODELING

As part of the methodology, we estimated a model to measure the correlation between the relative price variation and the change in the market share of countries. This correlation is

[5] Singletons are observations that stand alone within a fixed-effects category, with insufficient variation to contribute to the model estimation. In fixed-effects gravity models, they are removed because they do not help identify the coefficients and can lead to numerical problems or biases in the estimation.

what we call the price elasticity of the market share (β), and indicates how much the market share of a country varies, in terms of percentage, when there is a variation of 1% in its relative price. For the elasticity estimation, we used a gravity-model framework and estimated it following the methodology proposed by Silva and Tenreyro, 2006. In this section, we pass through the model built to estimate the elasticity, covering the gravity modelling approach, and why it was applied, the model specification, and the identification and robustness.

3.2.1 Gravity models

The gravity model has established itself as an important empirical approach for analysing international trade flows, due to its ability to consistently link trade to changes in transaction costs and market conditions. In its multiplicative specification, trade outcomes are modelled as a function of exporters' competitiveness aspects, destination market characteristics, and bilateral trade frictions, such as transport costs and tariffs (Head; Mayer, 2014). This framework is particularly well-suited for the present analysis, since the CBAM operates primarily through changes in the relative prices of imports among different exporters, affecting their competitiveness in the European market.

The main challenge in estimating gravity models lies in the use of the log-linear form via Ordinary Least Squares (OLS). As demonstrated by Silva and Tenreyro (2006), in the presence of heteroscedasticity, which is common in international trade data, the logarithmic transformation generates inconsistent estimators due to Jensen's inequality[6]. In such cases, the logarithmic transformation induces correlation between the error and the regressors, compromising the consistency of the OLS estimator. As an alternative, the authors propose the Poisson Pseudo-Maximum Likelihood (PPML) estimator, which directly estimates the multiplicative form of the model without requiring the logarithmic transformation of the dependent variable. In addition to being robust to heteroscedasticity, the PPML naturally accommodates observations with very low or zero trade flows. Recent literature has established the use of PPML combined with fixed effects as the primary empirical approach in gravity models applied to international trade. The inclusion of fixed effects is essential for controlling for various forms of unobserved heterogeneity that may simultaneously affect trade flows and trade costs, thereby reducing omitted variable problems and improving the identification of the coefficients of interest. Thus, the model retains the original multiplicative structure of the gravity equation while controlling for persistent structural characteristics and market-specific shocks (Correia; Guimarães; Zylkin, 2020; Yotov *et al.*, 2016).

Finally, although inspired by the gravity model estimated by PPML, which is widely used in the international trade literature, the model adopted in this study takes a different approach, focusing on the redistribution of market shares rather than directly on bilateral trade flows. This adaptation is appropriate because the CBAM operates primarily through relative cost differentials among exporters, affecting their competitive positioning in the European market. In this context, modelling market shares allows us to isolate the substitution effects associated with changes in relative prices, while preserving the multiplicative structure and robustness properties that underpin the use of the PPML estimator in the empirical literature on international trade.

3.2.2 The price elasticity of the market share model

As stated, this research approach aims to emulate the concept of price elasticity of demand, but in this case, focusing specifically on the substitution effects caused by the

[6] When under heteroskedasticity $E[\ln(y)] \neq \ln E(y)$, violating the exogeneity condition required for consistent OLS estimation in log-linearized models.

CBAM. Therefore, we use the market share as the dependent variable, rather than absolute volume, for two main considerations. First, import volumes are driven by many conditions other than competitive dynamics, such as energy prices, economic cycles, and internal industrial development, among many other factors. Second, the CBAM is a price-oriented policy that affects countries asymmetrically depending on their emission intensities, pushing for a shift in the distribution of EU imports. Market shares directly capture this redistributive dimension, making it the natural dependent variable for this analysis. Based on these considerations, the model was specified as follows:

$$s_{i,p,t} = exp(\beta_p \times ln(p_{i,p,t}) + \alpha_{i\times p} + \gamma_{t\times p}) \times \epsilon_{i,p,t} \quad (1)$$

Where $s_{i,p,t}$ is the market share of an exporter i in the market of the product p in a given year t. On the other hand, the explicative variable, $ln(p_{i,p,t})$, accounts for the log of relative price, constructed as the log-difference between exporter i effective price and a leave-one-out volume-weighted average of competitors:

$$ln(p_{i,p,t}) = ln[CIF_{i,p,t} \times (1 + \tau_{i,p,t})] - ln(\frac{\sum_{j\neq i}^{N_{p,t}} p_{j,p,t} \times q_{j,p,t}}{\sum_{j\neq i}^{N_{p,t}} q_{j,p,t}}) \quad (2)$$

In this case, exporter i effective price is calculated with the $CIF_{i,p,t}$ accounting for the product p price, considering freight and insurance, with an *ad valorem* import tariff $\tau_{i,p,t}$. The relative price is calculated by dividing the exporter i effective price by the volume-weighted average of all its competitors, so that the major exporters have a bigger impact on the world average price, approximating it to reality. Consistent with Head and Mayer (2014), who argue that quantity-weighted price indices better reflect the competitive conditions prevailing in the destination market. Before estimation, the relative price variable was winsorized at the top 1% of its distribution, replacing extreme values with the corresponding percentile threshold. This procedure limits the influence of outlier observations, typically associated with small or sporadic exporters, on the elasticity estimates, while preserving the distribution of the remaining observations (Hummels; Klenow, 2005).

Additionally, two fixed effects are used to absorb distinct sources of unobserved heterogeneity. The exporter-product fixed effect ($\alpha_{i\times p}$) is used to absorb time-invariant specific characteristics of an exporter, such as product quality or specialization, institutional relations that may impact commerce dynamics, structural cost advantages, and long-lasting differences in market power relations. The year-product fixed effect ($\gamma_{t\times p}$) aims to capture any punctual demand shocks of a given product p on a specific year t affecting all exporters. Both these fixed effects combined ensure that our calculated coefficient β identifies a competitive relationship between relative prices and market shares, free of structural exporter heterogeneity and aggregate demand conditions.

Finally, for the specific Brazilian and Argentinian price elasticity of the market share, we reestimated the model in the following format:

$$s_{i,p,t} = exp[(\beta_p + \delta_{p,c} \times 1_{i=c}) \times ln(p_{i,p,t}) + \alpha_{i\times p} + \gamma_{t\times p}] \times \epsilon_{i,p,t} \quad (3)$$

Where the $\delta_{p,c}$ is the differential term that captures the deviation of country c, in our case Brazil and Argentina, from the global elasticity. Therefore, the effective price elasticity of the market share for Brazil (β_{Br}) and Argentina (β_{Ar}) in a product p is $\beta_p + \delta_{p,c}$. To ensure $\beta_{p,c}$ precision, we applied a precision-weighted share of the country-specific differential $\delta_{p,c}$:

$$\beta_{p,c} = \beta_p + \frac{Var(\beta_p)}{Var(\beta_p) + Var(\delta_{p,c})} \times \delta_{p,c} \quad (4)$$

The weight assigned to the differential is determined by the relative precision of the two estimates: when $\delta_{p,c}$ is estimated with high uncertainty, as may occur for products where Brazil or Argentina have limited market presence, the weight converges to zero, and the simulation relies entirely on β_p. On the other hand, when the differential is precisely estimated, the country-specific differential receives substantially greater weight. This approach avoids arbitrary thresholds and allows the data to determine continuously how much country-specific information enters the simulation. This precision-weighted combination follows the shrinkage principle introduced by James and Stein (1992), whereby a unit-specific estimate with high variance is improved by borrowing strength from a more precisely estimated pooled counterpart. As formalised in the empirical Bayes framework of Morris (1983), the weight assigned to each component is determined by their relative precision: when the individual estimate is imprecise, the posterior converges toward the pooled estimator, and as precision increases, the individual estimate receives progressively greater weight. In the empirical panel literature, this bias-variance tradeoff between individual and pooled estimators is well established (Maddala *et al.*, 1997), and the continuous data-driven weighting avoids the discontinuities introduced by pre-testing or arbitrary significance thresholds for heterogeneity (Hjort; Claeskens, 2003).

3.2.3 Identification and robustness

Finally, a series of diagnostic tests was performed on the dataset to justify the choice of the PPML model and the specifications used. Table 2, showing the tests performed on the dataset, is provided below:

Table 2 – Panel diagnoses

Test	Aluminum	Cement	Iron & Steel
RESET (OLS log linear) p-value	0,001	0,001	0,001
RESET (PPML) p-value	0,414	0,294	0,637
Wooldridge F^2	2261.62 ***	4004.99 ***	502.97 ***
Hausman χ^2	0.30	96.17 ***	221.97 ***

*** p<0,01, ** p<0,05, * p<0,10. Standard errors clustered by exporter×product.
Source: Compiled from data obtained in the research.

Following Silva and Tenreyro (2006), we apply the Ramsey Regression Specification Error Test (RESET) to assess correct specification of the conditional expectation. The OLS log-linear model is rejected in all three sectors (p-value < 0.001), consistent with the heteroskedastic multiplicative errors that motivate the use of PPML. The PPML model is not rejected in any sector, confirming correct specification. The Wooldridge test (2002) rejects the null of no serial autocorrelation in all three sectors (F^2 significant at the 1% level). This is an expected outcome as market shares usually have strong time inertia, with market shares from

a year t being highly predicted by the level of the year before t_{-1}, due to aspects like supply long-term contracts, logistics, reputational advantages, and strong commercial relations. As a consequence, non-clustered standard errors would underestimate the real coefficient's variance, artificially inflating significance. Therefore, we used a $exporter\ x\ product$[7] cluster so that the relation between years observed by each pair of exporter i and product p is not considered independent.

The Hausman Test (1978) was conducted to understand the need for fixed effects in the model, instead of random effects, testing if the non-observed error for each panel unit ($exporter\ x\ product$) is correlated with the explanatory variable $ln(p_{i,p,t})$. The test rejects random effects in favor of fixed effects for iron and steel and cement (χ^2 significant at the 1% level), confirming that unobserved characteristics such as product quality and historical trade relationships are correlated with relative prices. For aluminium, the null is not rejected. Economically, this means that for this market, an exporter's structural market position does not systematically relate to its price competitiveness. Analysing the relative price histogram[8], for each sector, we can observe a higher concentration in the variation of aluminium relative prices to zero, indicating less price differential between exporters or years, which can be a possible cause for the test result. Nonetheless, the difference between fixed and random effects estimators is negligible ($\beta_{fixed\ effects}$ = -0.887 and $\beta_{random\ effects}$ = -0.886), so fixed effects are maintained across all sectors for consistency.

Additionally, in order to test the stability of β, four other alternative specifications were built using Claude Code (Annex B). The tested specifications are: (i) the OLS linear-log model, considering the natural logarithm to the dependent variable, aims to evaluate if the results are sensitive to the estimator choice; (ii) the PPML with a price lag of one period considers our base model but with relative prices of the year before, to understand if price effects on market shares are instantaneous or deferred; (iii) the PPML without the $product\ x\ year$ fixed effect aims to evaluate its contribution to the β identification; (iv) finally, we ran a placebo test that will be further explained after the table. The functional formats of alternative models are in Annex D, and the results can be seen in Table 3 below:

Table 3 – Alternative models and robustness[9]

Model	Aluminum (β)	Cement (β)	Iron & Steel (β)
PPML (base model)	−0,770 *** (0,227)	−1,309 *** (0,212)	−1,616 *** (0,394)
OLS ln(share)	−0,668 * (0,402)	−1,567 *** (0,258)	−1,617 *** (0,236)
PPML with price lag (t−1)	−0,328 (0,228)	−0,809 *** (0,194)	−1,067 *** (0,330)
PPML without $\gamma_{t\times p}$	−0,763*** (0,227)	−1,254*** (0,209)	−1,580*** (0,387)
Placebo test (t+1)			
β_1	−0,240 (0,148)	−0,202 (0,132)	−0,493 * (0,281)
β_0 reestimated	−0,720*** (0,227)	−1,223 *** (0,187)	−1,402 *** (0,352)
[β_1 participation in reestimated β_0]	-	-	35,16%

*** p<0,01, ** p<0,05, * p<0,10. Standard errors clustered by exporter×product.
Source: Compiled from data obtained in the research.

[7] Following Egger and Tarlea (2015), we verify that inference is robust to alternative clustering structures, including clustering by exporter and two-way clustering by unit and year, with standard errors varying by at most 13% and all coefficients remaining significant at the 1% level.

[8] Presented in annex C.

[9] In order to facilitate the analysis, the sector-level elasticity reported in the robustness table is a volume-weighted average of the product-level coefficients estimated individually, where each product's weight corresponds to its share of total EU imports within the sector over the base period. This aggregation ensures that the sector beta reflects the effective elasticity of the trade-weighted export portfolio.

Initially, we can observe that all β elasticities are negative, which indicates that increases in relative price lead to lower market shares, which is consistent with a normal good (Varian, 2014). In none of the 62 products analysed was the global β_p positive, indicating the non-existence of Giffen goods in the analysis. Furthermore, the results of the OLS ln(share) are similar in magnitude to those of the base model, but with higher variance and lower significance for aluminium, which reinforces the loss of efficiency already indicated by the RESET test and validates the choice of the PPML as the primary model. Robustness tests also show that the results are not extremely sensitive to changes in specification. The inclusion of lagged prices maintains significance in cement and iron and steel, suggesting slower adjustments in these sectors, while in aluminium the response is immediate. This has economic alignment with the more homogeneous relative prices observed for the aluminium sector, which could lead to more dynamic competition. The removal of the product-year fixed effects yields larger coefficients, indicating that these effects absorb common demand shocks and are relevant for the correct identification of elasticity.

The placebo falsification model includes the relative price of the year after $(t + 1)$ alongside the contemporaneous price in the baseline model, leading to two β, where β_0 captures the effect of the current relative price on market share and $\beta_1$is the lead term that captures the predictive power of the future price conditional on the current one. To do so, we lose one year (2025) of our sample. In order to isolate any temporal effect in comparison with the estimated β_1, we reestimate our base model considering the restricted sample, which is shown in the β_0 reestimated line. For this test, if the β_1 is small and not significant, we can interpret that the estimated relationship reflects contemporaneous competitive effects rather than spurious temporal correlations. The results indicate that the placebo coefficients remain substantially lower than the re-estimated contemporary coefficients in all sectors analysed. For aluminium and cement, the coefficients associated with the futures price are not statistically significant, suggesting the absence of a systematic relationship between futures prices and current market share after controlling for the contemporary price and the model's fixed effects. In the case of iron and steel, although the placebo coefficient is marginally significant, its magnitude remains considerably lower than the estimated contemporary effect, representing approximately 35.16% of the re-estimated principal elasticity. This suggests that the portion associated with the future price reflects a gradual adjustment of trade flows, which is expected as commercial relations between countries, which determine market shares, are not only driven by prices, but also by trade agreements, geographical aspects, bilateral relations, and other conditions that have low or no change over time.

Finally, although the model proved itself robust to alternative specifications, there are some important limitations. The first one is related to the temporal impact of relative price shifts on market share, which is considered only contemporaneous in the model. This means that the price elasticities of the market share are short-term elasticities, not considering slower and gradual changes that may occur in sectors with long-term contracts or higher switching costs. The second is that although we have a considerable level of granularity by analysis divided into HS4 and HS6 level codes, there is still a relevant level of heterogeneity between the products that compose those groups, which may introduce aggregation bias. That also means that the average price of those products of an exporter i can vary over time due to mix change rather than due to effective variations of the product's price, which can not be captured by our base model. And a third limitation relates to the potential endogeneity of the relative price variable. Although the inclusion of exporter-product and product-year fixed effects controls for time-invariant unobserved heterogeneity and common demand shocks, reverse causality cannot be fully ruled out: exporters losing market share may simultaneously adjust

their pricing strategy, creating a feedback from share to price that the fixed-effects structure does not absorb.

3.3 SIMULATION

The third step of this analysis was to use the estimated β to simulate the CBAM cost impact on the Brazilian and Argentinian competitiveness in the EU market. To do so, we constructed the price shock, using the emission factors, followed by the market share simulation, with the specific elasticity estimates in the modelling. Finally, we created an additional scenario by simulating the impact of the free trade agreement between Mercosur and the EU, combined with the CBAM.

3.3.1 The price shock construction

For the price shock, we estimated the additional cost that the CBAM would create for importers, changing the effective price paid for imported goods based on the emission factor default value combined with 3 scenarios of ETS prices. Following Regulation (EU) 2023/956, the CBAM additional import cost was estimated as the product of: (i) the embedded emissions associated with each product, measured in tonnes of CO_2 equivalent per tonne imported; and (ii) the corresponding CBAM certificate price, linked to the price of EU ETS allowances. Consistent with Article 9 of the Regulation, any effective carbon price already paid in the country of origin may be deducted from the adjustment. Therefore, the CBAM cost shall be:

$$CBAM\ Cost_{i,p,t} = ef_{i,p,t} \times ets_t \times (1 - cap_{i,p,t}) \quad (5)$$

Where $ef_{i,p}$ is the emission factor associated with the import of a product p from an exporter i. The ets_t stands for the EU ETS price that is the base for the CBAM certificate prices, which will be released by the European Commission quarterly in 2026 and weekly from 2027. In our methodology, we consider only one average ets_t yearly due to our yearly structure. Finally, the regulation defines that third countries with carbon markets established can deduct the amount already paid, defined as $cap_{i,p,t}$, from the final CBAM Cost to avoid duplicity on taxation. In our analysis, we do not consider this possible deduction due to lack of consistent and comparable data across countries. Therefore, the relative prices considering the CBAM Cost are:

$$ln(p^{cbam}_{i,p,t}) = ln[CIF_{i,p,t} \times (1 + \tau_{i,p,t}) + CBAM\ Cost_{i,p,t}] - ln\left(\frac{\sum_{j\neq i}^{N_{p,t}} [CIF_{j,p,t} \times (1+\tau_{j,p,t}) + CBAM\ Cost_{j,p,t}] \times q_{j,p,t}}{\sum_{j\neq i}^{N_{p,t}} q_{j,p,t}}\right) \quad (6)$$

In this case, we are considering that every exporter is affected by the CBAM according to its practical emission factor. We also consider the emission factor as zero for the exporters exempt from the policy, present in Annex III of Regulation (EU) 2023/956, and the EU countries. The CBAM price shock is estimated for:

$$\Delta ln(p^{cbam}_{i,p,t}) = ln(p^{cbam}_{i,p,t}) - ln(p_{i,p,t}) \quad (7)$$

Finally, for calculating the CBAM cost, we used three scenarios of ETS prices. The first scenario considers the ETS price of €75, following the first quarterly price release of 2026 by the European Commission (European Comission, 2026). The second and third scenarios are constructed mostly as sensitivity tests, but for anchoring them to real terms, they orbit around the forecast prices for 2027 and 2028. Therefore, the second and third scenarios consider the ETS prices of €95 and €115, respectively, anchored on forecasts from Reuters, 2026 and Altaghlibi and Gentile, 2025. These three price levels, €75, €95, and €115 per tonne of CO_2, define the range of CBAM cost scenarios applied in the market share simulation, allowing for the assessment of competitive impacts under both current and projected carbon price conditions.

3.3.2 Market share simulation

With the price shocks calculated, we can simulate the impact on market share based on our estimated β. The impact on the market share is described as:

$$\Delta s_{i,p,t}^{cbam} = s_{i,p,t} \times (exp[\beta_{i,p} \times \Delta ln(p_{i,p,t}^{cbam})] - 1) \quad (8)$$

Where the $\Delta s_{i,p,t}^{cbam}$ is the variation in the market share, in level terms. The $s_{i,p,t}$ is the base market share of an exporter i in the market of product p at year t. In this case, for both the $s_{i,p,t}$ and the $p_{i,p,t}$ we opted for using the average of the three last available years (2023, 2024, and 2025) to capture the structural trade pattern of the recent years, avoiding punctual shocks in price and volume. Finally, the $\beta_{i,p}$ is the price elasticity of the market share of a specific exporter i in the market of product p. The use of the exponential function is consistent with the functional form of the PPML, ensuring that the simulated market shares remain non-negative regardless of the magnitude of the price shock. Finally, since the CBAM affects exporters asymmetrically, simulated shares no longer sum to unity by construction. Shares are therefore normalized by product so that they sum to 100%, explicitly imposing the market-clearing identity consistent with the partial equilibrium setting of the analysis. This procedure is consistent with the framework of Yotov *et al.* (2016), which establishes that changes in trade costs trigger redistribution effects that must satisfy the importer's budget constraint.

Finally, we created a second triple of scenarios by considering the price effect of the EU-Mercosur free trade agreement. To do so, we considered, in the projection price, that for Mercosur exporters the import tariff applied $\tau_{i,p,t}$ would be equal to zero. This yields a total of six simulation scenarios, three carbon price levels combined with two tariff regimes (MFN and Mercosur), allowing for the assessment of how trade policy and climate policy interact in shaping the competitive positioning of Brazilian and Argentine exporters in the EU market.

4 RESULTS AND ANALYSIS

This section presents the empirical results of the study. To this end, it is divided into two parts: in the first, we seek to determine the price sensitivities of Brazil and Argentina in the European aluminum, iron and steel, and cement markets, and to what extent the estimated model is capable of explaining the allocation of market share among exporters. In the second

part, we use these estimates to simulate the impacts of the CBAM on the two countries' share of European imports, analyzing the expected effects by sector and carbon price scenario.

4.1 THE EU PRICE ELASTICITY OF THE MARKET SHARE

Before proceeding to the effective simulation, we should assess the model's capability to explain the market share patterns and anticipate the variation outside our sample. Moreover, we must understand the Brazilian and Argentine market share sensitivity by analysing their specific elasticities, which will help us understand the magnitude of CBAM impacts. We initiate by analysing the model's explanatory capacity in Table 4 below. The fit measure used is the square of the Pearson correlation coefficient between the predicted and observed market share values, here interpreted as R^2, consistent with the standard practice in the literature on gravitational models using PPML (Silva; Tenreyro, 2006).

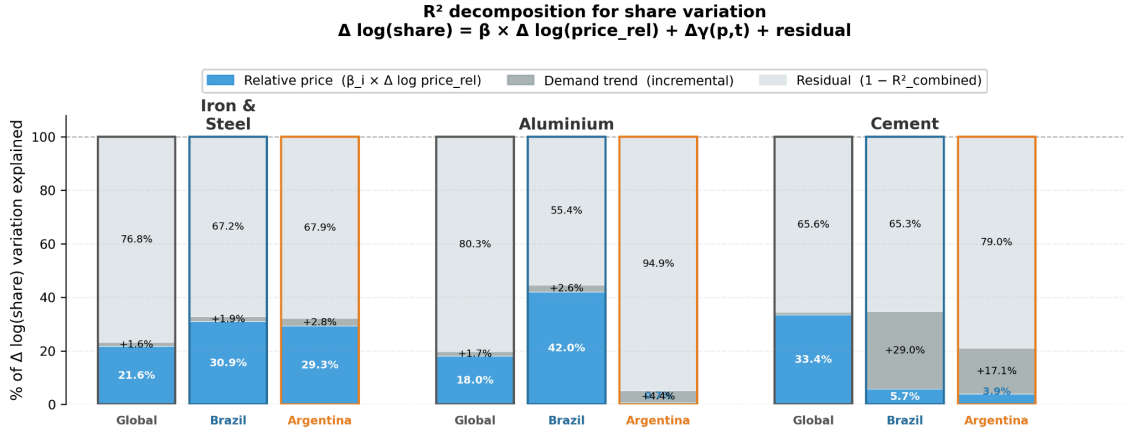
Table 4 – Base model share explanatory capacity

	Aluminum	**Cement**	**Iron & Steel**
R^2 (base model)	0,906	0,916	0,935
R^2 (fixed effects)	0,782	0,721	0,812
ΔR^2 price	0,123	0,195	0,123
% fixed effects	78,2%	72,1%	81,2%
% price	12,3%	19,5%	12,3%
% error	9,4%	8,4%	6,5%

Source: Compiled from data obtained in the research.

Table 4 shows that the base model achieves R^2 above 0.9 in all sectors, with fixed effects accounting for over 70% of explanatory power, reaching 81.2% in iron and steel, and relative price contributing a marginal 12.3% in aluminium and iron and steel and 19.5% in cement, consistent with the dominance of fixed effects in PPML models (Correia; Guimarães; Zylkin, 2020; Kwon *et al.*, 2026). This decomposition directly indicates the scope of CBAM's short-run impact: structural factors such as long-term bilateral relations, productive capacities, and quality premiums are unlikely to shift quickly, so the mechanism should initially operate through the marginal price channel (ΔR^2). Cement emerges as the most price-sensitive sector, consistent with its lower product differentiation relative to aluminium and iron and steel. To focus on short-run variation, we decompose R^2 into first differences of log market share, which eliminates the time-invariant exporter-product fixed effect by construction. The remaining variation is explained by two components: the relative price variation weighted by the estimated beta, and the common product-year demand trend. Results are presented in Figure 1.

Figure 1 - R² decomposition for share variation



Source: Compiled from data obtained in the research.

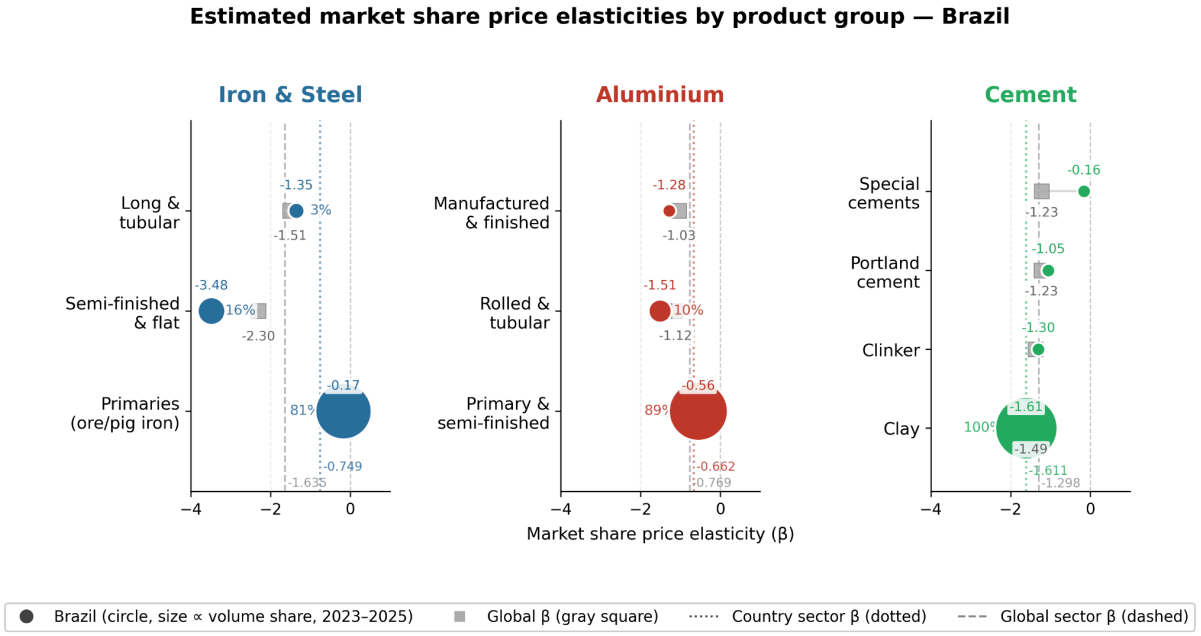
At the global level, relative price explains between 18% and 34% of the temporal variation in market share, with cement showing the strongest effect, consistent with its lower product differentiation, as discussed earlier. The common demand trend contributes marginally, adding 1 to 2 percentage points, confirming that aggregate market shocks are not the primary driver of market share redistribution among suppliers. The residual of 65% to 80% reflects idiosyncratic factors that vary among exporters over time without passing through the relative price channel, like supply shocks such as capacity expansions, exporter internal supply shocks, and long-term contracts, which the CBAM does not affect and which the simulation correctly holds constant.

For Brazil, the price signal is significant both in iron and steel (30.9%) and, above all, in aluminium (42.0%), the highest figure among all the pairs analysed, indicating that the mechanism of substitution via relative price operates in an empirically supported manner in these sectors. For Argentina, iron and steel shows a signal comparable to Brazil's (29.3%), which justifies its inclusion in the simulation. Argentine aluminium, however, has a price R^2 of only 0.7%, with 94.9% of the share variation determined by factors unrelated to relative price, which indicates that the model probably will not be able to generate consistent outcomes with the simulation. The poor model reliability with the Argentine aluminium sector, which accounted for 81,3% of total exports, in euros, of the CBAM-affected sector, is the main loss regarding the contributions of this work. The reason for the unpredictability of the Argentine share by our method is the monopolised aluminium market, in which Aluar Aluminio Argentino SAIC represents nearly all exports (Misirlian; Barcia, 2018). In this case, specific strategic Aluar decisions may have influenced the Argentinian sample and year-over-year variations. Therefore, the presence of a near-monopoly exporter introduces idiosyncratic firm-level effects that are difficult to capture with a model based primarily on sectoral and market-wide determinants. Lastly, for cement, the price signal is less than 6% for both countries, while the demand trend outweighs the price component. The low significance of relative price in the cement market reflects distinct dynamics: for Brazil, a secondary supplier (around 1% of the EU market), there is a structural decline in market share while the EU market grew, indicating displacement by more competitive competitors rather than a price response; for Argentina, the presence is merely residual (≈0%), without constituting a continuous market. These results directly underpin the analytical scope decisions adopted: the CBAM simulation is conducted for iron and steel from Brazil and Argentina and for aluminium from Brazil, pairs in which the relative price mechanism demonstrates empirical relevance for explaining competitive dynamics in the European market. Cement and

Argentine aluminium are excluded from the simulation due to the absence of the mechanism that the CBAM triggers, especially in the short-run.

That stated, we can initialize by analysing our elasticities. To facilitate the analysis, we separated each sector into three or four product categories, which go from less to more complex goods[10]. We start by analysing the Brazilian grouped elasticities, which can be seen in Figure 2:

Figure 2 - price elasticities of market share for Brazil



Source: Compiled from data obtained in the research.

Starting with the iron and steel sector, we can see that Brazilian exports were highly concentrated in the primary products, accounting for more than 81% of total CBAM-affected iron exported, followed by semi-finished and flat products with 15%. For the primary products, we can see that the Brazilian market elasticity is virtually equal to the global, and both are very inelastic, considering β = -0,17, which means that a variation of 1% of relative prices would lead to a negative variation of 0,17%. These lower calculated elasticities are probably due to primary iron products having their prices mainly defined in the stock market, which means there is low variation between exporters, and time variations tend to be symmetric among exporters, therefore being captured by our fixed effects. The same would be true about the primary and semi-finished aluminium group, which also has a β lower than one. Back to the iron and steel sector, we can see that for the semi-finished and flat iron and steel products, Brazil has more price sensitivity than the average exporter, having a market share elasticity 1.5 times higher than the global average. Therefore, it is expected that CBAM should have its impacts amplified for Brazil for the semi-finished and flat iron and steel products, as Brazilian exports are more price sensitive than competitors. In this case, the impact of small CBAM costs for Brazil can be equalised to the impact of higher CBAM costs for their competitor, which indicates that Brazilian exports can be more sensitive to the CBAM in this group.

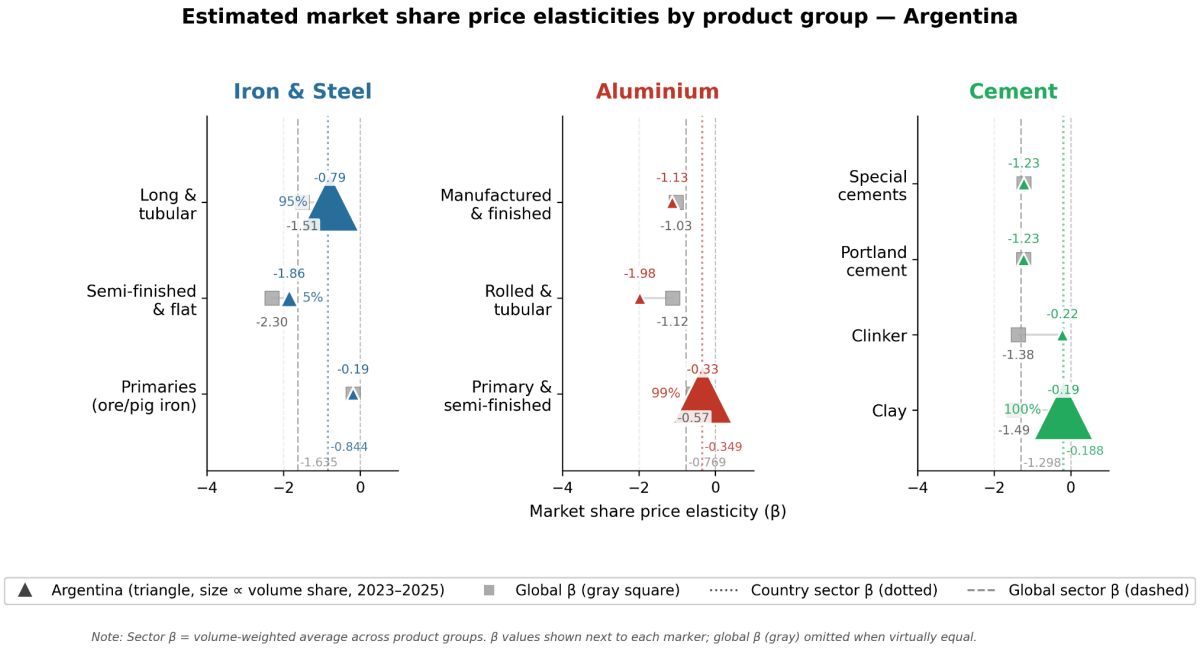
In the case of aluminium, we can see even more concentrated exports in the primary and semi-finished products from 2023 to 2025, with this group accounting for 89% of

[10] The category of each product can be seen in Annex E.

Brazilian CBAM-affected aluminium products. For the primary products, the same as in iron and steel primary products occurs, Brazilian elasticities being virtually equal to the global elasticity. In the aluminium case, we can observe lower variability between Brazilian and global elasticity, yet for both rolled and tubular, and manufactured and finished groups, the average exporter is less price sensitive than Brazilians. Finally, for the cement sector, only clay was exported from 2023 to 2025, with global clay being slightly less price-sensitive than the Brazilian. As a result, we can see that Brazilian exporters compete mainly in the primary good markets, especially in the iron and steel and aluminium sectors, while the intermediate complexity levels are secondary. From a CBAM perspective, this positioning offers a natural hedge: primary goods carry lower embedded emissions, reducing the carbon charge itself, while their lower price elasticities limit the market share response to any residual cost wedge.

For Argentina, the export composition is considerably different, especially in the iron and steel sector. The Argentine market share elasticities can be seen in Figure 3 below:

Figure 3 - price elasticities of market share for Argentina



Note: Sector β = volume-weighted average across product groups. β values shown next to each marker; global β (gray) omitted when virtually equal.

Source: Compiled from data obtained in the research.

The first main difference lies in the iron and steel sector, as Argentine CBAM-affected exports are concentrated in the long and tubular group, which is the most complex category. However, unlike Brazil, Argentina's main export group is less price sensitive than the global average, being almost half, which can mean an advantage, as more complex goods tend to have higher emission factors. For aluminium, we can see even more concentrated exports in the primary products, accounting for 99% of Argentine exports. Also in that case, the Argentine elasticity is lower than the global, meaning that Argentina should have less impact. Summarizing, both Brazil and Argentina show lower sector elasticity, especially in the iron and steel sector, but also in aluminium, with a smaller difference. Regarding the iron and steel sector, in Brazil's case, the lower sector elasticity comes from the majority of primary goods that showed less price sensitivity, in relation to the base EU market. In the case of Argentina, the lower sector elasticity comes from the difference in elasticities for the long and tubular products. This configuration may mitigate the CBAM impacts for Brazil and Argentina.

Finally, as a structural validation exercise, we used the definitive anti-dumping duties imposed by the EU on imports of cold-rolled flat steel from China (+22.1%) and Russia

(+36.1%), established by Regulation (EU) 2016/1329, as a quasi-natural experiment to test the model's ability to reproduce market share redistributions induced by exogenous trade cost shocks. The products covered by the measure and present in our database are HS codes 7209, 7211, 7225, and 7226, all segments of flat steel of high commercial relevance. The underlying mechanism is structurally analogous to that of the CBAM: in both cases, an additional cost is imposed on importers from specific countries, raising their relative price in the European market and generating substitution pressure in favor of unaffected exporters. The procedure exactly replicated the CBAM simulation methodology: the PPML model was estimated exclusively using data from the 2010 to 2015 period, ensuring that the parameters did not incorporate information about the shock; the anti-dumping tariff was translated into a relative price change using the same formula used in constructing the carbon shock; and the simulated market share was obtained using the redistribution equation applied to the 2015 base period, with a test for the year 2017, the first year in which the definitive measures took full effect.

The results, evaluated exclusively for the countries directly affected by the tariffs, indicate that the model using global betas estimated during the pre-shock period achieved a Pearson correlation of 0.902 between the simulated and observed magnitudes, correctly predicting the direction of the change in 7 out of 8 cases. When the estimation was replicated using country-specific betas for China and Russia, following the same methodology applied to Brazil and Argentina in the CBAM simulation, the correlation rose to 0.951. This dual result has a direct methodological implication: the performance of the global betas validates the substitution mechanism captured by the model as structurally robust, while the improvement obtained with country-specific betas confirms that the estimation of elasticity differentials by exporters adds significant predictive accuracy and justifies their use in the CBAM simulation for Brazil and Argentina. A graphic analysis of the results can be seen in Annex F.

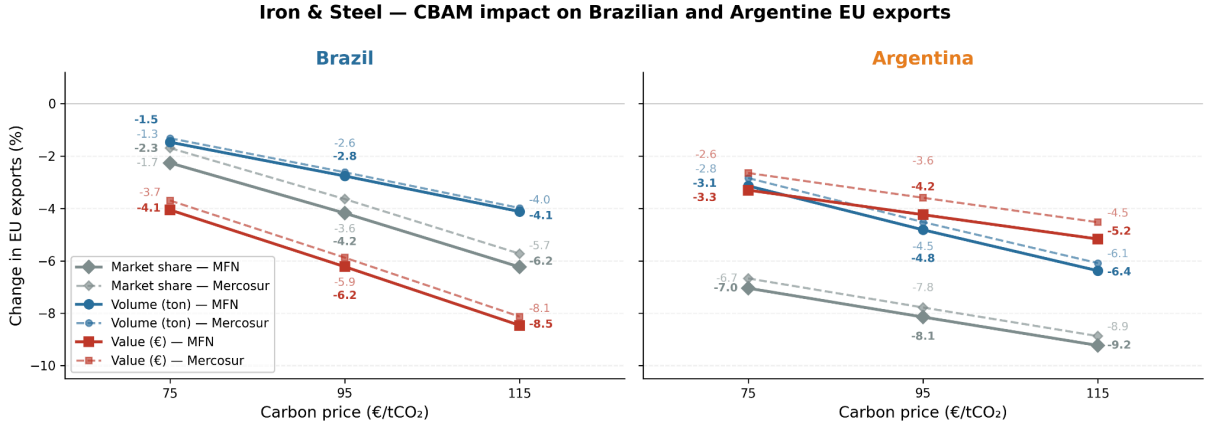
4.2 THE CBAM SIMULATIONS

This section will cover the simulations of the CBAM impact in the iron and steel sector for Brazil and Argentina, and for the aluminium sector for Brazil, as these combinations of sector and exporter showed the most reliability. The simulations were conducted as stated in subsection 3.3.

4.2.1 Iron and Steel

The iron and steel sector is the most relevant CBAM-affected sector for Brazil, given that between 2023 and 2025, Brazilian iron and steel exports to the EU reached €4,6 billion, accounting for 89,2% of the total analysed sectors. For Argentina, on the other hand, iron and steel accounted for 18,7% of the CBAM-affected sector's exports, with less than 25 kilotons exported during the period, compared with 12,923 kilotons exported by Brazil. The estimated impact can be seen in Figure 4 below. The MFN scenario, represented by the continuous lines and darker numbers, considers full import tariffs and the CBAM impact. The Mercosur scenario, represented by the dashed lines and lighter numbers, represents the impact scenarios in which Brazil and Argentina have zero import tariffs, but are still impacted by the CBAM. The volume impact in tons exported, represented by the blue lines, considers no variation in total imported volume to the EU, being calculated through the share variation. Lastly, the economic value impact, represented by the red lines, considers the cash amount that would enter Brazil and Argentina, that is, the CIF value, calculated from the estimated volumes and the CIF price of products between 2023 and 2025. All scenarios are estimated with the 2026 emission factors.

Figure 4 - Impact scenarios for Brazil and Argentina exports in the iron and steel sector



Source: Compiled from data obtained in the research.

For Brazil, the impact observed on export values (a decline of 3.7% to 8.5% as the carbon price rises from €75 to €115) is consistent with the findings of Feijó and Kuik (2016), who calculated a significantly larger decline of 15.1% in the scenario involving direct emissions from the production process. This discrepancy is expected, given that Feijó's study uses the GTAP-EP general equilibrium model, which captures both the substitution effect and the income effect (contraction in total demand) in an integrated manner, while this simulation focuses exclusively on the substitution effect via changes in market share. In contrast, the simulated reduction in Brazil's export volume (between 1.5% and 4.1%) is consistent with the findings of Mortha *et al.* (2023). Using a structural gravity model, the authors estimated a sectoral decline of 2.24% for the South American region under a carbon price of USD 87, suggesting that models based on trade costs and multilateral resistance tend to more closely reflect the market share dynamics captured in this analysis.

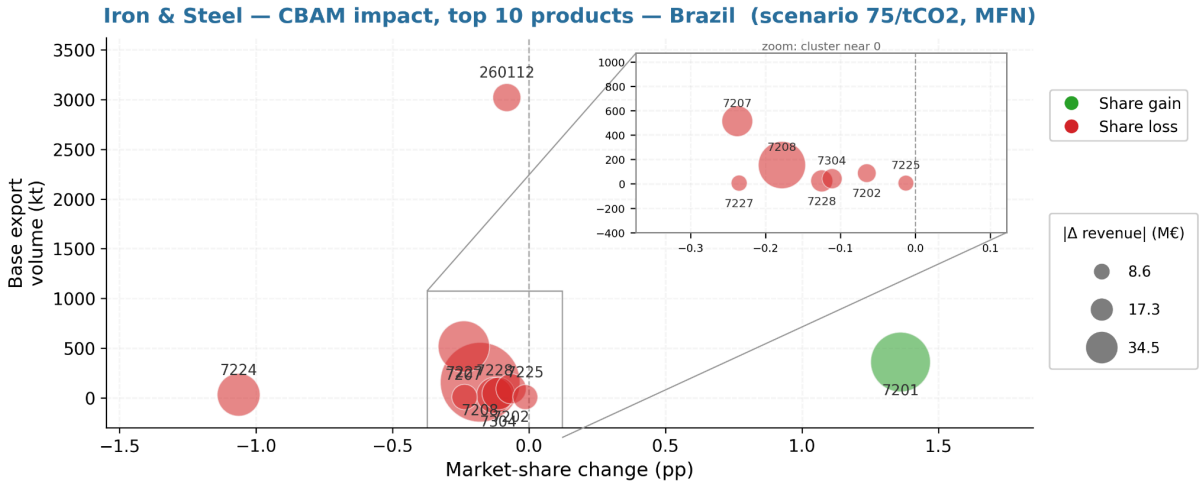
Concerning Argentina, the loss of market share, which was considerably larger than Brazil's (a 9.2% drop versus 6.2% at €115), is consistent with the competitive logic presented by Perdana, Vielle and Oliveira (2023). This is likely due to the lower carbon intensity of Brazilian production, which gives the country a relative competitive advantage over neighbors with more emissions-intensive production processes, resulting in lower tariff "vulnerability" under CBAM rules. In this context, it was found that Brazil's average emission factor is 9.3% lower than Argentina's. And when considering the product mix, with emission factors weighted by the value of exports to the EU, the difference is even greater, with Brazil's emission factor being 30.8% lower than Argentina's. Additionally, the trend toward modest negative impacts on trade flows is consistent with the findings of Korpar, Larch and Stöllinger (2022), who estimated marginal declines of 0.21% in total Brazilian exports and 0.05% in Argentine exports under the European mechanism. If we recalculate the impacts on the iron sector, we can see that when considering the entire sector (including products not affected by the CBAM) the economic impact decreases from -4.1% to -1.9% in the €75 scenario for Brazil, while Argentina sees a reduction from -3.3% to -2.8%, under the MFN scenario. However, if we consider the impact on total exports from both countries, the impact is -0.14% for Brazil and -0.009% for Argentina.

Regarding the Mercosur scenarios, the damping effect observed in this simulation (dotted lines) is consistent with the methodological assumptions of Korpar, Larch and Stöllinger (2022). The authors demonstrate that, in gravity models, the existence of regional agreements and pre-existing tariff barriers acts as a buffer, reducing the relative weight of the new carbon tariff on the total cost of imports and preserving the competitiveness of bloc

members vis-à-vis global exporters that do not have preferential access. In this context, both Brazil and Argentina already have relatively low import tariffs, averaging 0.84% and 0.73%, respectively, on a value-weighted basis. Thus, when we convert the CBAM into an equivalent *ad valorem* tariff, we find rates of 13.11% for Brazil and 7.97% for Argentina, also weighted by export value, which suggests that the difference in agreement scenarios would not have such a significant impact in this case. Lastly, an analysis of the slopes of the impact curves reveals that Brazil exhibits a higher sensitivity to carbon price increases compared to Argentina. In the MFN scenario, as the price rises from €75 to €115, the impact on Brazil's export value deepens by 4.4p.p., compared with 1.9p.p. for Argentina. This steeper decline suggests that Brazilian products face higher trade elasticities of substitution, indicating they are more susceptible to being replaced by domestic European production or other competitors as the CBAM levy scales up. This is consistent with the market share elasticities weighted by export volume, in which Brazil shows an elasticity 1.54 times higher than Argentina's.

At last, a notable difference observed in Figure 4 is where the greatest impact lies for each country. In the case of Brazil, we can see that the economic impact is considerably greater than the impact on market share and volume, while in Argentina the economic impact appears smaller than the impact on market share and volume, depending on the scenario. To explain what causes this divergence, we break down the impacts by exported product in the case of Brazil, as shown in Figure 5 below.

Figure 5 - Estimated impact by iron and steel product for Brazil



Source: Compiled from data obtained in the research.

First, we can see that the main product in terms of export volume, iron ores and concentrates (260112), has a moderate impact in terms of value, since it has a lower price and lower price elasticity, as shown in Figure 2. On the other hand, the main impact lies in higher-value-added products, namely flat-rolled products of iron or non-alloy steel (7208) and semi-finished products of iron or non-alloy steel (7207), which have highest market share elasticity. In this regard, for iron ores and concentrates, Brazil holds a 10.2% share of EU imports between 2023 and 2025, based on export volume, ranking fifth. Nevertheless, among the top 5 largest exporters, Brazil has the highest emission factor, while two European

countries have a zero emission factor: Sweden and the Netherlands[11], with market shares of 17.1% and 15.7%, respectively.

Nevertheless, although iron ores and concentrates account for 70.1% of Brazil's export volume to the EU, they represent only 27.6% of the export value in euros, indicating lower prices and resulting in a relatively small economic impact compared to other products in Brazil's export mix. In this regard, it is rolled and semi-finished products (7208 and 7207), which together account for 25% of Brazil's exports to the EU by value, that will drive estimates of the decline in Brazilian exports. In the case of semi-finished products (7207), Brazil is the sixth-largest exporter to the EU with a 4.9% share, and despite having the lowest emission factor among non-European countries in the top 10, Germany and France, which together account for 17.4%, are exempt from CBAM. In the case of rolled products (7208), however, Brazil does not even appear in the top 10, where the top 6 countries are exempt from CBAM. Thus, for both cases, it was considered that CBAM is likely to increase intra-block trade in more intermediate products with higher value added, which could create upward pressure on demand for more primary products such as iron ore and pig iron.

Thus, although we saw earlier that primary products, as exemplified by iron ore, were not expected to have a significant economic impact, even when excluding intra-block exports, the case of pig iron (7201) appears to be the exception. This is the only case where Brazil showed a gain in market share and a positive change in export value, which is due to the fact that all of the top four exporters, which hold 66.6% of the market, are non-European countries. In this regard, Brazil is the third-largest exporter and has an emissions factor 50% lower than its main competitors. The gain in this product aligns with the hypothesis of increased intra-EU exports of more industrialized goods, which would require greater demand for primary products from relatively cleaner countries, such as Brazil. Finally, we can see that in the Brazilian case, products with intermediate value added are the most affected, mainly due to the increase in intra-EU trade, which leads to an economic impact on Brazil greater than the impact in terms of volume and market share.

In the case of Argentina, on the other hand, the main product affected by the CBAM is seamless tubes, pipes, and hollow profiles made of iron (excluding cast iron) or steel (7304). In this case, between 2023 and 2025, this product accounted for 72.5% of Argentina's exports to the EU in terms of volume, while it accounted for 71.9% in terms of export value. Consequently, the discrepancy between physical and economic impact depends on the distribution of the trade pattern across products with different profiles. In Brazil, where volume and value are driven by different products (cheaper ore and more expensive semi-finished goods), the two metrics diverge. In Argentina, where a single product (7304) accounts for about 72% of both, volume and value are more closely correlated. The same product decomposition for Argentina can be seen in Annex G.

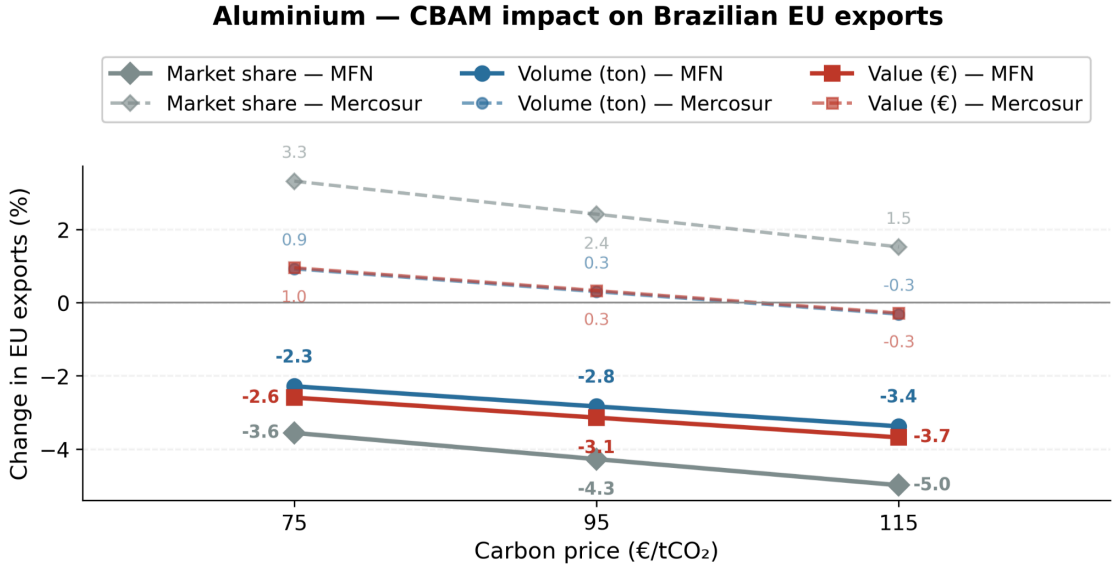
4.2.2 Aluminium

The aluminum sector accounted for approximately 7% of the total Brazilian exports of CBAM-affected sectors to the EU between 2023 and 2025. In this case, CBAM coverage is nearly complete, with 94.6% of Brazilian exports during the base analysis period being affected by CBAM. Nevertheless, in total terms, the portion of the aluminum sector affected by CBAM represents only 0.27% of trade between Brazil and the EU. Finally, in the

[11] As a robustness check, the model was re-estimated excluding the Netherlands, a major EU hub unlikely to be a direct iron ore producer, and subsequently excluding all EU exporters, to assess whether re-exports might be artificially suppressing the CBAM cost for intra-EU flows. The exclusion of the Netherlands alone produced no change; excluding all EU exporters shifted Brazil's iron ores volume impact from -0.8% to +0.1%, suggesting re-exports have no material effect on the results

aluminum sector, the Mercosur scenario plays a more important role, as can be seen in Figure 6 below:

Figure 6 - Impact scenarios for Brazil exports in the aluminium sector



Source: Compiled from data obtained in the research.

For aluminium, the simulated reduction in export volume under MFN (2.3% to 3.4%) is again smaller than the 14.1% contraction projected by Feijó and Kuik (2016), for the same methodological reasons discussed in the iron and steel case, and more closely aligned with the 0.57% decline estimated by Mortha *et al.* (2023) for non-ferrous metals in South America. Yet, in the aluminum case, the most intriguing results reside in the difference between the base scenario and the zero tariff Mercosur scenario. In both volume and value variations, in the case of the free trade agreement, the reduction of the import tariffs can compensate for the CBAM impact for ETS prices of €75 to €95. This result stems from a structural peculiarity of the aluminum sector that distinguishes it from the iron and steel sector. Unlike the steel industry, where the applied import tariff is practically zero (0.84% on a value-weighted average), aluminum faces a substantial MFN tariff of around 5.15%. The *ad valorem* equivalent of the CBAM cost, in turn, stands at 5.69% in the €75 scenario, rising to 7.21% and 8.73% in the €95 and €115 scenarios. Therefore, a break-even point is observed around €68/tCO_2[12], below which the elimination of the tariff provided by the agreement more than compensates, in terms of absolute cost, for the burden introduced by the CBAM. Since the first two carbon price scenarios (€75 and €95) operate close to this range, the Mercosur tariff reduction is sufficient to neutralize, and even reverse, the competitive effect of the CBAM.

It is worth noting that the gain in market share observed under the Mercosur regime persists even in scenarios where the absolute cost of the CBAM exceeds the eliminated tariff. This occurs because the impact on market share is determined by the relative price, not the absolute cost: as the CBAM also burdens other suppliers subject to the mechanism, and Brazil has a lower emission factor than its direct competitors, being 16% lower than CBAM-affected countries, the relative increase in costs from competing sources preserves part of Brazil's competitive advantage. Finally, the aluminium sector for Brazil has a similar structure to the iron and steel sector for Argentina, with only one product, unwrought aluminium, representing around 82% of total traded value. In this case, Brazil is not a major player in the EU market, representing only 0,35% of the EU imported value of unwrought aluminum.

[12] Graphical analysis comparing the import tariff and equivalent *ad valoren* CBAM cost available in Annex H.

5 FINAL CONSIDERATIONS

This study aimed to analyze the potential effects of the EU's Carbon Border Adjustment Mechanism (CBAM) on the relative competitiveness of Brazilian and Argentine exports in the European market, based on the estimation of market share price elasticities and the simulation of relative price variations induced by the mechanism in the aluminum, iron and steel, and cement sectors.

The main results indicate that the impacts of the CBAM on Brazil and Argentina are negative, but moderate, and heterogeneously distributed among countries, sectors, and products. For Brazil, the iron and steel sector concentrates the greatest exposure to the mechanism, with losses in market share and export value that increase as the price of carbon rises. This impact is disproportionately concentrated in higher value-added products, mainly semi-finished and flat products, while primary products show less sensitivity to the mechanism, with pig iron being the only product in which Brazil projects a gain in market share, due to its lower relative emission intensity compared to competitors. For Argentina, the distinct export structure, concentrated on long and tubular products in iron and steel, with substitution elasticities below the global average, implies a comparatively lower exposure in this sector. A noteworthy result for both countries is the potential role of the Mercosur-EU agreement: in Brazilian aluminum, the tariff elimination foreseen by the agreement would be sufficient to fully offset the cost of the CBAM in the lowest carbon price scenarios, reversing the negative impact into a gain in market share.

From the point of view of contributions, the study advances on three fronts. Theoretically, it directly operationalizes the redistributive transmission mechanism of the CBAM impact in relative prices among competing exporters, complementing a literature that mostly captures aggregate macroeconomic effects. Empirically, the product-level disaggregation at the HS code level allows identification of which products within each sector concentrate vulnerability and which offer more resilient competitive positions, proving decisive in explaining why value and volume impacts diverge within each country's portfolio. Practically, the results offer inputs for trade policy by identifying exposure by product and quantifying the extent to which the Mercosur-EU agreement can offset the carbon burden. The central implication is that sector-level analyses systematically obscure the heterogeneity that determines the actual competitive impact on each country's export portfolio: emission intensity and substitution elasticity vary substantially across products within the same sector, making product-level disaggregation a necessary step for accurate exposure assessment.

Yet, some important limitations arise. The estimated elasticities are short-term in nature, not incorporating gradual adjustments resulting from long-term contracts or productive restructurings, which implies that the simulated impacts may underestimate the cumulative effects over time. Additionally, the absence of deductions for domestic carbon pricing may overestimate the effective costs for exporters with developing carbon pricing mechanisms. Based on these limitations, directions for future research are suggested. A natural advancement would be to deepen the disaggregation of the analysis, operating directly at the CN8 level, the nomenclature used by the CBAM regulation itself for emission factors, eliminating the residual aggregation bias still present and capturing heterogeneities within product groups currently treated as homogeneous. Replicating the methodology for a broader set of exporting countries would allow for systematic comparisons of exposure to the mechanism and make the results more generalizable. Finally, as data from the definitive period become available from 2026 onwards, it will be possible to transform this prospective analysis into an ex-post assessment, refining the estimates based on real exogenous price variations induced by the policy.

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ANNEX A – APPROVAL SHEET

LORENZO VIANA LIMA

CBAM AND EXPORT COMPETITIVENESS: MARKET SHARE ELASTICITY ESTIMATES AND SIMULATIONS FOR BRAZIL AND ARGENTINA

Final project submitted to the Undergraduate Program in Economic Sciences at the Faculty of Economic Sciences of UFRGS, as a partial requirement for obtaining the Bachelor's degree in Economics.

Approved in: Porto Alegre, June 25, 2026.

EXAMINING BOARD:

Prof. Dr. Flávio Tosi Feijó – Advisor

UFRGS

Prof. Dr. André Moreira Cunha

UFRGS

Dra. Thais Diniz Oliveira

UNDP

ANNEX B – DESCRIPTIVE TABLE ON THE USE OF AI

Table 5 - Descriptive table of AI use in the workplace

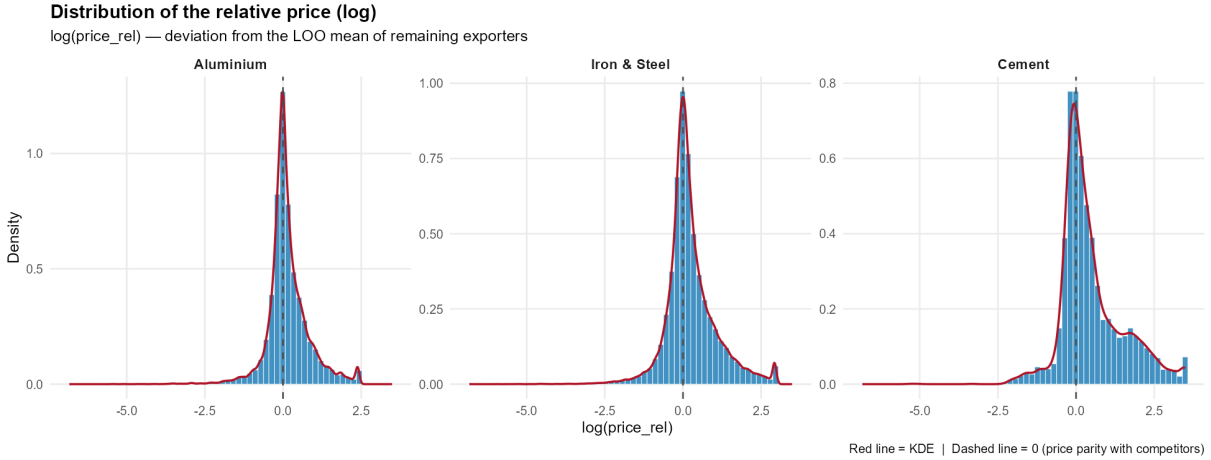
Methodological stage	Name of the tool used	Date	Objective	Prompts
References gathering	Consensus	19/02/2026	References gathering	Please gather me articles related to the following topics: - CBAM impact on Brazil and Argentina; - Gravity models for trade.
Methodology	Claude Ai	02/04/2026	Building alternative specification models	Please, based on the model (.text) attached, build alternative models in R language, with functional formats… (see in Annex D).
Revision	Claude Ai	31/05/2026	Writing revision	Please perform a writing revision of my paper. Look for grammatical and semantic improvement points.

Source: own elaboration.

Model based in:
CENTRO UNIVERSITÁRIO SENAI CIMATEC. **Guia para uso de IA Generativa no Centro Universitário SENAI CIMATEC**. Salvador, fev. 2024. Available at: https://www.universidadesenaicimatec.edu.br/wp-content/uploads/2024/03/GUIA-DE-IA-NA-EDUCACAO.pdf. Accessed: 05 jun. 2026.

ANNEX C – RELATIVE PRICE HISTOGRAM

Figure 7 - Distribution of relative prices



Source: built with data from the research.

ANNEX D – ALTERNATIVE MODELS FUNCTIONAL FORMATS

Table 6 - Alternative models

Model	Functional format
OLS log(share)	$ln(s_{i,p,t}) = \beta_p \times ln(p_{i,p,t}) + \alpha_{i\times p} + \gamma_{t\times p}) \times \epsilon_{i,p,t}$
PPML with price lag (t−1)	$E[s_{i,p,t}\|X] = exp(\beta_p \times ln(p_{i,p,t-1}) + \alpha_{i\times p} + \gamma_{t\times p}) \times \epsilon_{i,p,t}$
PPML without $\gamma_{t\times p}$	$E[s_{i,p,t}\|X] = exp(\beta_p \times ln(p_{i,p,t}) + \alpha_{i\times p}) \times \epsilon_{i,p,t}$
Placebo test (t+1)	$E[s_{i,p,t}\|X] = exp(\beta_{p,0} \times ln(p_{i,p,t}) + \beta_{p,1} \times ln(p_{i,p,t+1}) + \alpha_{i\times p} + \gamma_{t\times p}) \times \epsilon_{i,p,t}$

Source: built with data from the research.

ANNEX E – PRODUCT GROUPS

Table 7 - HS Codes analysed and groups

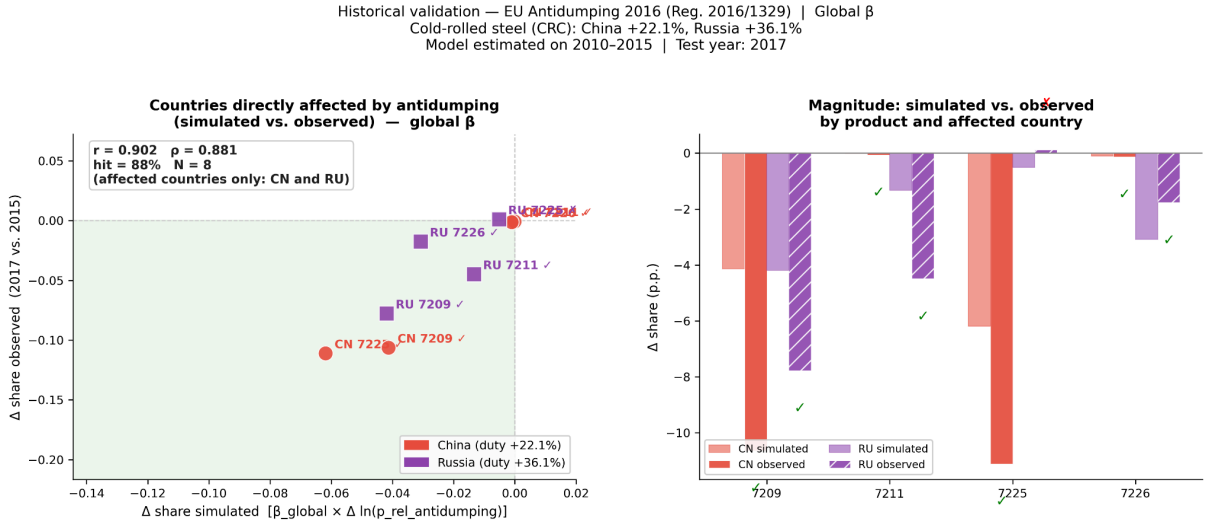
Sector	HS Code	Description	Group
Iron & Steel	260112	Agglomerated iron ores and concentrates	1 – Primaries (ore & pig iron)
Iron & Steel	7201	Pig iron and spiegeleisen	1 – Primaries (ore & pig iron)
Iron & Steel	7202	Ferro-alloys (ferro-manganese etc.)	1 – Primaries (ore & pig iron)
Iron & Steel	7203	Ferrous products by direct reduction of iron ore	1 – Primaries (ore & pig iron)
Iron & Steel	7205	Granules and powders of pig iron or steel	1 – Primaries (ore & pig iron)
Iron & Steel	7206	Ingots of iron and non-alloy steel	2 – Semi-finished & flat
Iron & Steel	7207	Semi-finished products of non-alloy steel	2 – Semi-finished & flat
Iron & Steel	7208–7212	Flat-rolled products of non-alloy steel	2 – Semi-finished & flat
Iron & Steel	7218–7221	Stainless steel — ingots, flat-rolled, bars	2 – Semi-finished & flat
Iron & Steel	7225–7226	Flat-rolled products of silicon-electrical/other alloy steel	2 – Semi-finished & flat
Iron & Steel	7213–7217	Bars, rods, angles, wire of non-alloy steel	3 – Long & tubular
Iron & Steel	7222–7229	Bars, wire, ingots of stainless/alloy/high-speed steel	3 – Long & tubular
Iron & Steel	7301–7302	Sheet piling, railway track materials	3 – Long & tubular
Iron & Steel	730300	Cast iron tubes and pipes	3 – Long & tubular
Iron & Steel	7304–7306	Line pipe, tubes and pipes (seamless and welded)	3 – Long & tubular
Iron & Steel	7307–7308	Pipe fittings, structures (bridges, towers etc.)	3 – Long & tubular
Iron & Steel	730900 –731100	Tanks, reservoirs, containers for compressed gas	3 – Long & tubular
Iron & Steel	7318	Screws, bolts, nuts	3 – Long & tubular
Iron & Steel	7326	Grinding balls and similar articles	3 – Long & tubular
Aluminium	7601	Unwrought aluminium	1 – Primary & semi-finished

Aluminium	7603	Aluminium powders and flakes	1 – Primary & semi-finished
Aluminium	7604	Aluminium bars and rods	1 – Primary & semi-finished
Aluminium	7605–7607	Wire, plates/sheets/strip, foil	2 – Rolled & tubular
Aluminium	7608 –760900	Tubes, pipes and fittings	2 – Rolled & tubular
Aluminium	7610–7612	Doors/windows, tanks/drums/cans	3 – Manufactured & finished
Aluminium	761100– 761300	Reservoirs, containers for gas	3 – Manufactured & finished
Aluminium	7614	Stranded wire and cables	3 – Manufactured & finished
Aluminium	7616	Nails, staples, other articles	3 – Manufactured & finished
Cement	252310	White clinker	1 – Clinker
Cement	252321	White Portland cement	2 – Portland cement
Cement	252329	Grey Portland cement	2 – Portland cement
Cement	252330	Aluminous cement	3 – Special cements
Cement	252390	White hydraulic cements (other)	3 – Special cements
Cement	250700	Calcined clay	4 – Clay

Source: built with data from the research.

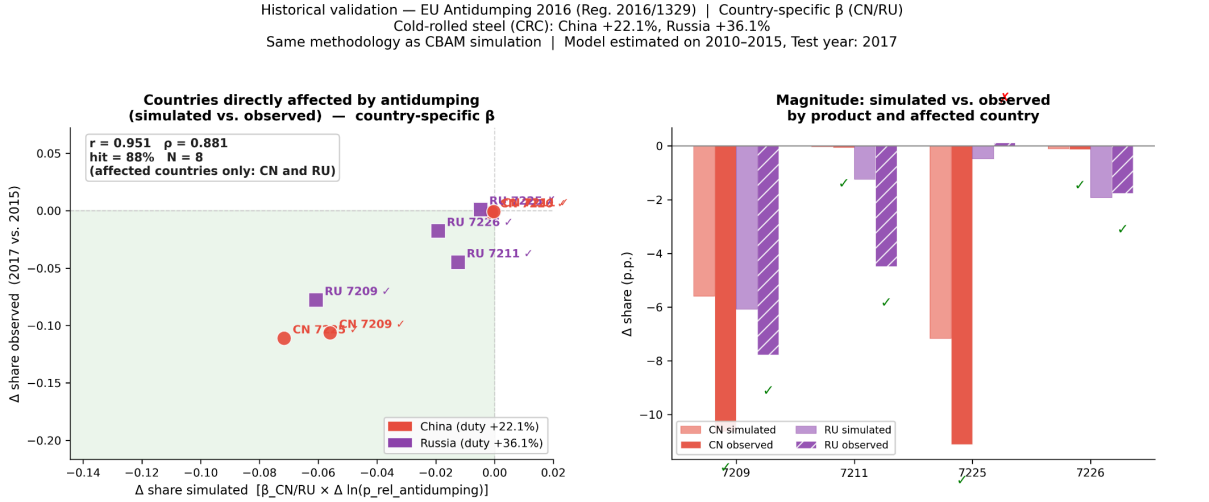
ANNEX F – GRAPHIC ANALYSIS OF THE HISTORICAL VALIDATION TEST

Figure 8 - Historical validation using the global market-share elasticities



Source: built with data from the research.

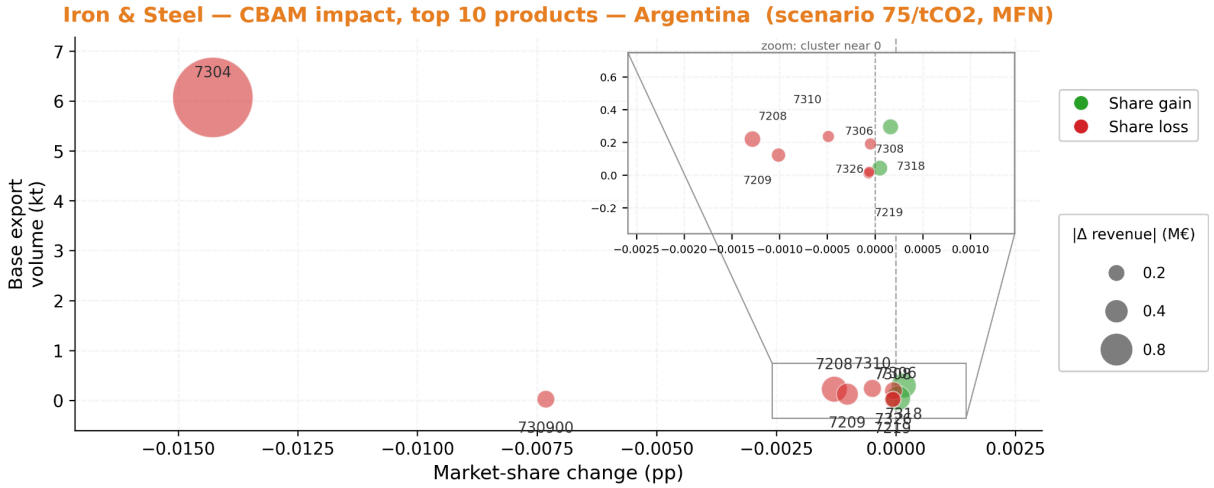
Figure 9 - Historical validation using the country-specific market-share elasticities



Source: built with data from the research.

ANNEX G – ESTIMATED IMPACT BY IRON AND STEEL PRODUCT FOR ARGENTINA

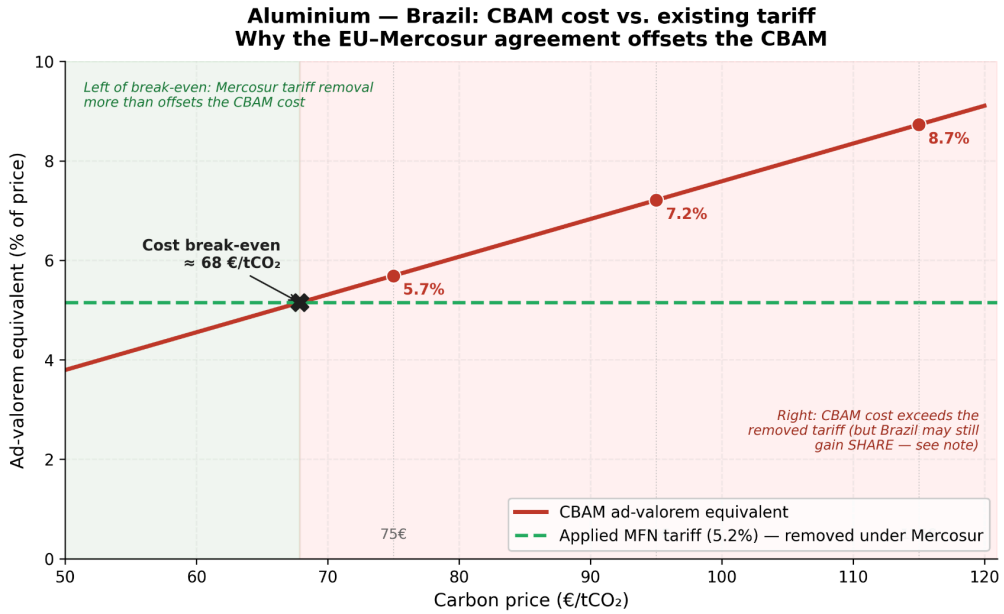
Figure 10 - Estimated impact by iron and steel product for Argentina



Source: built with data from the research.

ANNEX H – IMPORTS TARFIS AND CBAM COST COMPARISON

Figure 11 - Imports tariffs and CBAM cost comparison for the Brazil aluminium sector



Note: the cost break-even (~68 €/t) is where CBAM = tariff in absolute terms. Brazil keeps a SHARE gain beyond it because the CBAM also raises rival prices (carbon-intensity advantage).

Source: built with data from the research.